San Diego Builds Space-Radiation Monitoring Infrastructure—While Training the Workforce to Run It

Local nonprofit's Sentinel-Bio+ constellation protects GPS, communications, and defense systems from space weather—built largely by students

When GPS falters, satellite links degrade, or spacecraft fail early, the invisible culprit is usually space radiation—intense fluxes of charged particles driven by solar storms and trapped in Earth's magnetosphere.

For insurers pricing billion-dollar space assets, airlines routing flights, and defense contractors operating global systems, that unpredictability translates into real cost.

The San Diego Space Institute (SDSI), a nonprofit focused on applied space systems development, is deploying Sentinel-Bio+—a constellation of small satellites that continuously measures space radiation fluxes and their biological effects in low Earth orbit. The goal: deliver the persistent, high-quality data that satellite operators, insurers, and government agencies need to predict and manage space-environment risk.

"This isn't an academic exercise," said Sean Casey, president of SDSI. "The constellation is designed to operate under the same constraints and standards as commercial systems."

Why Now

The Sun is nearing the peak of Solar Cycle 25, and the number of satellites in orbit has exploded. More assets in space means dramatically higher aggregate exposure—and rising operational risk when things go wrong.

San Diego is getting ahead of the problem not with multibillion-dollar flagship missions, but with practical, repeatable, low-cost infrastructure. The planned constellation will consist of multiple small satellites launched on rideshare missions over the next three to five years.

The System

Sentinel-Bio+ places multiple small satellites in sun-synchronous low Earth orbit, providing frequent global coverage—far superior to the sporadic passes of a single large observatory.

Think many thermometers instead of one weather station.

The resulting data feeds radiation model validation, commercial risk assessments, and public-sector resilience planning. The focus is utility, not novelty.

The program also engages local suppliers, software developers, launch providers, and systems integrators involved in each mission cycle.

The Workforce Pipeline

What sets this apart from typical student projects: undergraduates and graduate students from San Diego State, San José State, and other regional universities work inside genuine industry workflows—formal design reviews, requirements tracking, interface control documents, qualification testing, verification, and mission operations.

Students participate in requirements definition, subsystem testing, documentation, and operations planning under professional supervision.

This is not a club. It is a structured, mentored pipeline that mirrors real aerospace programs. Graduates enter the workforce with documented systems-engineering experience on flight hardware—and many stay in San Diego.

"We're building mission-qualified engineers, not just graduates," Casey said. "The difference matters to regional employers."

Regional Advantage

San Diego already has deep strength in defense, aerospace, software, and biotech. Sentinel-Bio+ sits at the intersection of those sectors, leveraging university partnerships and a dense employer base that relies on resilient space infrastructure.

Instead of exporting talent, the region is growing its own.

Cost Discipline

Credibility requires restraint. Sentinel-Bio+ is deliberately low-cost, rideshare-launched, and incrementally scalable. Small satellites and frequent launches limit risk: failures are survivable, lessons are incorporated quickly, and the constellation improves with each flight.

The approach reflects how space infrastructure is increasingly being developed—through smaller, repeatable systems rather than single flagship missions.

Governance

As a nonprofit steward, SDSI ensures long-term continuity beyond any single student cohort. Education, research, and operational missions are kept distinct, giving partners, donors, and regulators clear lines of accountability.

What Success Looks Like

In three to five years: a fully populated constellation delivering operational data, dozens of alumni placed in local aerospace and defense roles, and follow-on missions funded by demonstrated value.

The emphasis is on operational milestones—satellites launched, data streams validated, workforce pipeline established—rather than aspirational goals.

No hype. Just infrastructure that works—and a trained workforce ready to keep building it.